

Lab 4 – VibeCoding

Arm Wrestle Showdown – Chat History / Development Log

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Project	53_arm_wrestle
GitHub Repository	https://github.com/sankgit14/PES1UG24AM416-Vishwas-

1. Objective

Use Vibe Coding to repair the provided Arm Wrestle Showdown game, implement the specified features one at a time, test each change, and maintain separate Git commits for the tasks.

2. Task 1 – Fix Inverted Arm Push

Problem: The original LEFT/RIGHT player inputs moved the arm toward the computer. **Change:** Both player push operations were changed from `self.arm_position += 4.2` to `self.arm_position -= 4.2`. **Testing:** The player was able to push toward the player side and win. **Commit:** Fix inverted arm push direction.

3. Task 2 – Dynamic AI Surge Cycle

Change: The AI was given NORMAL, SURGE, and COOLDOWN states with timed transitions. SURGE increases AI force and COOLDOWN reduces it before returning to NORMAL. The AI state is reset when the game is restarted. During testing, an accidental PLAYING-state typo was identified and corrected. **Commit:** Add dynamic AI surge cycle.

4. Task 3 – AI Surge and Exhaustion Indicators

Change: A visible AI SURGE! indicator was added during the AI SURGE state. An exhaustion state variable and EXHAUSTED! indicator were also added. The exhaustion behavior was tested while preserving the original stamina recovery and AI balance. **Commit:** Add AI surge and exhaustion indicators.

5. Task 4 – Counter-Surge Bonus

Change: When AI transitions from SURGE to COOLDOWN, a 1.5-second counter-surge window is activated. During this window, an alternating player push receives double push strength (4.2 to 8.4) and a temporary stamina recovery bonus of 4.0. **Testing:** The feature was tested immediately after the surge ended, and the stronger counter push and stamina bonus were observed. **Commit:** Add counter-surge bonus.

6. Git Commit History

Stage	Commit message	Status
Starter	Add Lab 4 arm wrestle starter project	Pushed
Task 1	Fix inverted arm push direction	Pushed
Task 2	Add dynamic AI surge cycle	Pushed
Task 3	Add AI surge and exhaustion indicators	Pushed
Task 4	Add counter-surge bonus	Pushed
Housekeeping	Ignore generated files and videos	Pushed

7. Video Evidence

The submission includes a before-change recording and an after-change recording. They are stored locally in the project under `videos/before.mp4` and `videos/after.mp4`.

8. Final Repository State

The repository was pushed to the personal GitHub repository. A `.gitignore` file was added so generated Python cache files and the local videos folder are not accidentally committed.